Introduction

1. Context & Rationale

The Cognitive Cost of Screen Media

Children spend 5.5 to 8.6 hours daily on entertainment media during the critical developmental window when executive function matures. Passive consumption and unconstrained generative AI threaten to replace the mental exertion required to build problem-solving stamina.

Entertainment Surge

Daily entertainment screen use increased by **+17% across two years**. Tweens (8-12) average 5h 33m daily, while teens (13-18) consume 8h 39m outside schoolwork.

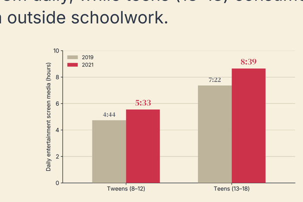

FIG 1. Daily screen media by age group, 2019 vs 2021 (+17% growth).

The Critical Window

Complex executive function develops rapidly between **ages 7 and 14** (rising 50% to >90%), overlapping with smartphone ownership surge (50%+ by age 11).

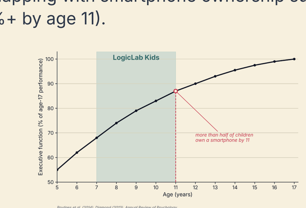

FIG 2. Executive function trajectory by age (CAS complex-EF, ages 5-17).

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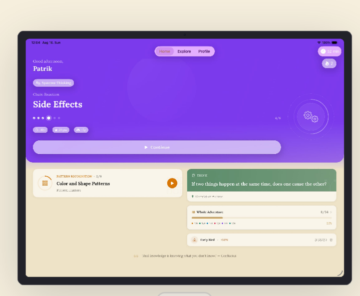

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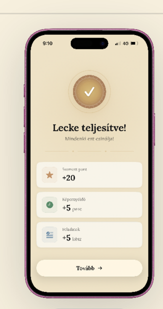

ANDROID

NATIVE SWIFTUI & KOTLIN COMPOSE

OS-ENFORCED REWARD INCENTIVE LOOP

The Premack Behavioral Contingency Model

Under Premack's Principle (1959), high-probability behaviors (screen entertainment) effectively reinforce low-probability behaviors (structured logical reasoning). LogicLab Kids establishes an unbreakable OS-level contingency where screen time is earned through independent problem-solving.

REFERENCES

[1] Common Sense Media (2019, 2021). Media Use by Tweens and Teens.

[2] Brydges et al. (2014). Executive function trajectory across ages 5-17 (N=2,036).

[3] Premack, D. (1959). Toward empirical behavior laws: I. Positive reinforcement.

[4] Rideout et al. (2022). The Common Sense Census: Media use by kids.